

MASTER PITCH & ENGINEERING DOSSIER

DOCURE

Autonomous Clinical Intelligence, NVIDIA AI Pathology OCR, Spatial Medical Radar, and Medication Adherence Engine.

 DUAL-ENGINE OCR

Mozilla PDF.js text extraction + Meta Llama 70B & 90B Vision on NVIDIA NIM.

 SPATIAL RADAR

TomTom Places API v2 + Leaflet GIS with 1-click doctor calling & routing.

 SMART ADHERENCE

Firestore cloud sync, 20s polling daemon & Web Audio frequency chimes.

PROJECT: DOCURE Health Intelligence System
TEAM: Debug-Creww (Hackathon 2026)

RELEASE: Production v2.4.0
LOCATION: Greater Noida / NCR

01 Executive Problem Statement & The 2 AM Healthcare Crisis

Across urban and semi-urban populations, **over 70% of households** experience severe anxiety and delay when dealing with medical uncertainty. The traditional patient journey is broken across four critical friction points:

1. Pathology Report Incomprehensibility

Diagnostic lab reports (CBC, Lipid, LFT, KFT) contain dozens of technical biomarkers (MCV, TIBC, SGOT, HbA1c). Patients resort to unguided internet searches, triggering unnecessary panic or ignoring genuine critical markers.

2. Emergency Location & Verification Friction

During sudden night emergencies, public search maps often display outdated numbers, closed clinics, or police stations instead of verified 24/7 trauma centers with direct emergency room lines.

3. Medication Non-Adherence Epidemic

According to the WHO, **50% of chronic patients** fail to take medicines correctly. Traditional reminder apps require manual alarm setups and lack cloud synchronization across caregiver devices.

4. Triage Delay Before Doctor Visit

Patients either over-triage mild conditions to crowded ERs or under-triage life-threatening cardiac symptoms, causing dangerous delays in emergency care.

The DOCURE Thesis

DOCURE closes this complete loop: It extracts lab reports into plain-language clinical insights, maps verified local doctors & 24/7 pharmacies, enforces daily pill schedules via background daemons, and provides 1-tap emergency sirens with GPS dispatch.

02 The 3-Minute Hackathon Pitch Script (Spoken Transcripts)

0:00 - 0:45 • THE HOOK

"Good morning respected judges! How many times have you or your family received a blood test report, Googled the complex parameters at 2 AM, and panicked because the internet told you it's a fatal illness?"

Today, millions struggle with **three healthcare gaps**: incomprehensible lab reports, emergency location friction, and missed daily medications. To solve this entire loop, we engineered **DOCURE**."

0:45 - 2:15 • PRODUCT WALKTHROUGH

"**DOCURE is an autonomous Clinical AI Healthcare Assistant.**

- **First, our Lab Report Analyzer:** Upload any PDF or photo from Dr Lal or SRL. Mozilla PDF.js extracts lossless text, and Meta Llama 70B & 90B Vision on NVIDIA NIM parse every test parameter sequentially — categorizing them into HIGH, NORMAL, and LOW with a Clinical Health Score and diet plan.
- **Second, our Real-Time Medical Radar:** Using TomTom POI Geosearch and GPS, DOCURE scans your surrounding 10 km radius for verified Hospitals, Clinics, Diagnostic Labs, and 24/7 Pharmacies with **1-Click Direct Calling** and Google Maps routing.
- **Third, our Medication Compliance Daemon:** Synced via Firebase Firestore, a 20-second background worker triggers in-app toast alerts, Web Audio chimes, and 1-click Google Calendar alarms.
- **Fourth, Emergency SOS:** Synthesizes a browser-native 850Hz ambulance siren and dispatches live GPS coordinates to family contacts in 10 seconds."

2:15 - 3:00 • PUNCHLINE & CLOSE

"DOCURE does not replace doctors — it empowers patients with instant clarity and connects them to the right care at the right time. With DOCURE, healthcare is instant, verified, and right in your pocket. Thank you!"

03 Node-by-Node Technical Feature Breakdown

 NODE 04: NVIDIA NIM AI Pathology & Lab Report OCR

Dual-Engine Architecture: If the patient uploads a digital PDF (e.g. Dr Lal PathLabs, SRL, Thyrocare), **Mozilla PDF.js** reconstructs lossless text coordinates directly in the browser and transmits them to **Meta Llama 3.1 70B Instruct**. If a camera photo is uploaded, **Meta Llama 3.2 90B Vision** parses the physical image.

```
PDF Upload → PDF.js Binary Extractor → Llama 3.1 70B (via /api/nvidia Proxy) → Balanced JSON Parser
Output: [Serial Test Rows] + [Unit & Range] + [HIGH/LOW Flags] + [Health Score: 0-100] + [Specialist Referral]
```

 NODE 03: Real-Time Spatial Medical Radar & GIS

Integrates **TomTom Search & POI API v2** with category codes: **7321** (Hospitals), **7322** (Doctors), **7324** (Clinics), and **7326** (Pharmacies). Renders hardware-accelerated **Leaflet.js** pulsing map pins with driving times, distance, and 1-click calling.

 NODE 02: Medication Adherence Daemon

Cloud persistence powered by **Firebase Firestore** under `users/{uid}/reminders` with real-time websocket snapshot listeners. An autonomous **20-second polling daemon** monitors medication schedules, generating Web Audio multi-frequency chimes and action toast overlays.

 NODE 01: Symptom Triage Chat

Multi-LLM triage desk with voice input (`webkitSpeechRecognition`) and speech synthesis (`window.speechSynthesis`) supporting Hindi & English.

 NODE 05: Emergency SOS Siren

Generates dual-frequency 850Hz/650Hz siren waves using browser `AudioContext.createOscillator()` with 10s auto-dispatch.

04 Complete Tech Stack Matrix & Architecture

Layer	Technology / Library	Exact Purpose & Engineering Benefit
Frontend Core	React 18.2 + Vite 5.4	Virtual DOM reconciliation, fast HMR dev server, Rollup production bundling.
Styling System	Tailwind CSS 3.4	Custom mint-sage palette, glassmorphism, responsive CSS grid and animations.
AI Inference	NVIDIA NIM Cloud API	Enterprise-grade LLM execution for Meta Llama 70B & 90B Vision models.
Reverse Proxy	Vite HTTP Proxy	Routes <code>/api/nvidia</code> to <code>integrate.api.nvidia.com</code> to prevent browser CORS blocks.
Spatial Mapping	Leaflet 1.9 + TomTom API	Interactive map tiles, custom pulsing DivIcons, category POI queries.
Cloud Database	Firebase Firestore v10	Real-time NoSQL synchronization for prescription reminders and patient profiles.
Audio Synthesis	Web Audio API	Browser-native oscillator chime and ambulance siren generation (zero MP3 assets).

05

Competitive Landscape & Business Monetization

Feature Dimension	DOCURE	Practo / 1mg	ChatGPT / WebMD
Pathology OCR Extraction	Lossless PDF.js + 70B/90B Dual AI	Manual upload / Text only	Generic summary without tabular sorting
Biomarker Status High/Low	Exact reference ranges & color cards	Not available in chat	Hallucinates ranges frequently
Live GIS Facility Radar	Real-time TomTom GPS + 1-Click Call	Sponsored doctor listings only	No real-time maps or GPS
Autonomous Medication Daemon	20s polling listener + Web Audio	Basic push notification	None
Client-Side Privacy	Local processing, ephemeral AI payload	Centralized patient database	Cloud chat history stored

 **B2B Lab Partnerships**

Direct API integration with diagnostic chains (Dr Lal, SRL) to automatically send smart patient explainer summaries.

 **Hospital SaaS Portal**

Verified listing badges and emergency ER dispatch routing for local hospital networks.

 **Pharmacy Integrations**

1-click prescription fulfillment and refill reminders via local partner pharmacies.

06

Master Judge Q&A Defense Directory

Judge's Tough Question	Winning Direct Technical & Clinical Answer
Q1: Is the AI diagnosing diseases? What about liability?	"DOCURE does not diagnose. It performs triage, biomarker classification against printed normal reference ranges, and educational interpretation. Every report analysis directs the patient to the appropriate medical specialist."
Q2: Why use both 70B text and 90B vision models?	"Digital lab PDFs contain structured text. Extracting text directly with PDF.js and feeding it to Llama 3.1 70B is 10x faster, cheaper, and 100% immune to image blur. 90B Vision is reserved for camera photos."
Q3: How do you prevent hallucinated biomarker values?	"Our balanced JSON parser enforces sequential matching against the raw text vectors extracted by PDF.js. If a test value is not present in the source report, the AI is prompted to omit it rather than guess."
Q4: How does the background medicine daemon work if the app is minimized?	"The notification scheduler runs an active 20-second polling loop coupled with Firestore snapshot listeners and supports 1-click sync with native Google Calendar alarms for system-level alerts."
Q5: How is patient data secured?	"Lab reports are processed ephemerally through authenticated NVIDIA NIM endpoints without persistent public storage. Reminder documents are stored in user-isolated Firestore collections."